Fast and Calibrated: Amortized Simulation-Based Inference for Log-Gaussian Cox Processes

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Abstract

Log-Gaussian Cox processes (LGCPs) are the default model for aggregated spatial count and point-pattern data in ecology, epidemiology and the social sciences, but Bayesian inference for their parameters is costly: every dataset requires a fresh Markov chain Monte Carlo (MCMC) run that must repeatedly integrate over a high-dimensional latent Gaussian field. *Amortized* simulation-based inference (SBI) removes this per-dataset cost while preserving the full Bayesian posterior, and recent work has shown it is feasible for LGCPs; the open question is whether the resulting posteriors can be *trusted*. We answer it with a systematic calibration audit. We train a neural posterior estimator—a convolutional summary network feeding a masked autoregressive flow—on simulations from a grid-discretized LGCP, and evaluate it with the field's most demanding diagnostics. The learned posterior passes simulation-based calibration on all three parameters (baseline intensity, spatial variance and correlation range) and attains near-nominal credible-interval coverage. The common point-estimate "regression" approach, by contrast, fails calibration in a way that only simulation-based calibration exposes: its intervals are right on average yet uninformative per dataset—their width barely varies and does not track the actual error—so they cannot flag which estimates to trust. Against a gold-standard No-U-Turn Sampler, the amortized posteriors are statistically indistinguishable—a classifier two-sample test accuracy of 0.543 (chance = 0.5)—yet are produced in 12.67 ms per dataset versus 280 s for MCMC, a per-dataset speed-up of $\sim 22,123\times$ that repays its one-time training cost after only 5 datasets. Going beyond global parameters, we also amortize the *intensity surface* itself: a single network returns a structured-Gaussian posterior over the whole latent field whose low-rank correlation term keeps not only pointwise intervals but *aggregate* functionals (total abundance, regional counts) calibrated—matching a gold-standard field posterior at a fraction of the cost. Finally, we exploit the likelihood-free nature of the approach to go beyond the standard model: under two non-standard observation processes—*preferential sampling* (informative locations, where the usual analysis is biased) and *spatial aggregation* (only coarse regional totals observed, the change-of-support problem)—adding the process to the simulator yields calibrated, bias-corrected inference that a design-aware estimator delivers at no extra methodological cost, and that correctly widens its uncertainty as aggregation destroys information. On a real crime point pattern the estimators—trained only on simulations— reproduce a NUTS analysis (parameters and intensity map) with no retraining, in seconds rather than minutes. Amortized SBI thus makes calibrated LGCP inference not only fast but extensible to observation processes that defeat likelihood-based tools.

1 Introduction

Spatial point patterns—the locations of trees, disease cases, crimes, or animal sightings—are ubiquitous, and the *log-Gaussian Cox process* (LGCP) is their canonical model [Møller et al., 1998, Diggle et al., 2013]. An LGCP is a doubly stochastic Poisson process whose log-intensity surface is a Gaussian process (GP); its parameters—overall abundance, the strength of spatial aggregation, and the spatial scale over which that aggregation operates—are exactly the quantities ecologists and epidemiologists want to estimate and compare across regions, species or time periods.

Inference is the bottleneck. The LGCP likelihood is intractable because it marginalizes a latent GP over the whole domain; in practice one discretizes the domain into a fine grid and treats the per-cell log-intensities as a high-dimensional latent Gaussian vector [Møller et al., 1998]. Fully Bayesian inference then proceeds by MCMC over this vector jointly with the covariance parameters, which is slow and delicate because the Gaussian field and the covariance hyperparameters are strongly coupled [Christensen et al., 2006, Murray et al., 2010]. Deterministic approximations such as INLA with an SPDE representation [Rue et al., 2009, Lindgren et al., 2011] accelerate this enormously but are approximate, tied to Gaussian-Markov structure, and still solved *per dataset*. In modern applications—disease mapping across thousands of administrative units, species distribution models over many taxa, or streaming crime data—the same model is fit again and again to structurally identical but numerically distinct datasets. Paying a full inference cost every time is wasteful.

Amortized simulation-based inference offers a different trade [Cranmer et al., 2020, Zammit-Mangion et al., 2025]. One trains a neural conditional density estimator $q_\phi(\boldsymbol{\theta} \mid \mathbf{x})$ on pairs $(\boldsymbol{\theta}, \mathbf{x})$ drawn from the prior and the simulator, so that a single forward pass returns an approximate posterior for any new dataset $\mathbf{x}$. The cost of inference is moved *up front* and *shared* across all future datasets, and the method never evaluates the likelihood—only simulates from it—so it extends naturally to observation processes (thinning, preferential sampling, aggregation) that break analytic approaches.

This idea has already reached the LGCP: Wang et al. [2026] apply the BayesFlow amortized estimator [Radev et al., 2020] to one- and two-dimensional LGCPs and validate it against MCMC, and a broader line of work uses neural networks to estimate spatial point-process parameters as point predictions [Vihrs, 2022]. Feasibility, in other words, is established. The question we take up is the next one, and the one that decides whether such estimators can be *used*: are the amortized posteriors *trustworthy*? Concretely—do they pass the field's strongest calibration diagnostics across the whole prior, not just agree with MCMC on a handful of cases; how do they compare with the prevailing point-estimate paradigm; what does the choice of summary statistics cost in calibration; and how do they behave when the simulator is misspecified? And, once trust is established, we ask what amortization makes newly *possible*: reconstructing the entire intensity surface, and calibrated inference under observation processes—preferential sampling and change of support—that standard likelihoods handle poorly, before closing on a real point pattern. This paper is a systematic answer.

Contributions. Building on the feasibility established by prior amortized LGCP work [Wang et al., 2026], we contribute both a rigorous *trust audit* and several *new capabilities*, using a transparent, dependency-light neural posterior estimator (a convolutional summary network feeding a masked autoregressive flow; Section 3):

1. We subject the learned posterior to the strongest calibration diagnostics in the SBI toolkit—simulation-based calibration over the full prior, plus marginal coverage and an *adaptive-uncertainty* analysis—and show it passes on all three parameters (Section 4).

Table 1: Overview of experiments. Four of the seven are validated head-to-head against a gold-standard No-U-Turn Sampler (NUTS).

Experiment (§)	Inferential target	Validated against	Headline result
Calibration (4)	parameters	prior (SBC)	passes SBC; regressor fails
MCMC benchmark (5)	parameters	NUTS	C2ST 0.543, $22,123\times$ faster
Intensity surface (6)	latent field	NUTS	map corr. 0.990; aggregates calibrated
Misspecification (7)	parameters	—	graceful, diagnosable degradation
Preferential samp. (8)	params + γ	NUTS	unbiased; corrects biased baseline
Change of support (9)	parameters	NUTS	calibrated; uncertainty tracks info. loss
Real data (10)	params + field	NUTS	zero-shot; map corr. 0.993

2. We run a controlled comparison against the prevailing point-estimate paradigm [Vihrs, 2022]: a regression network matches the NPE on point-estimate accuracy yet is badly miscalibrated in a way that marginal coverage alone would miss, because its uncertainty does not adapt to the data (Section 4). We further ablate learned versus hand-crafted spatial summary statistics.

3. We benchmark against a gold-standard No-U-Turn Sampler in two dimensions: the amortized posteriors are statistically indistinguishable from MCMC (classifier two-sample accuracy 0.543) at a $\sim 22,123\times$ lower per-dataset cost, breaking even after 5 datasets (Section 5).

4. We amortize the **intensity surface**, not just the parameters: a U-Net emits a structured (low-rank + diagonal) Gaussian posterior over the whole latent field in one forward pass. It is calibrated pointwise *and* for aggregate functionals—where a per-cell posterior fails—and matches a gold-standard NUTS field posterior (Section 6).

5. We report an honest robustness study under kernel misspecification, showing graceful rather than catastrophic degradation (Section 7).

6. **A methodological extension** that goes beyond prior amortized LGCP work: inference under *preferential sampling*. Because the informative design is just part of the simulator, a design-aware NPE corrects the bias that the standard ignorable-design analysis suffers, stays calibrated, and recovers the preferentiality parameter—validated against a joint-model NUTS reference (Section 8). A second observation model, *spatial aggregation* (change of support), is handled the same way: inference from coarse regional totals stays calibrated and widens its uncertainty to match the information lost (Section 9).

7. Finally, we apply the estimators—trained *only* on simulations—to a real crime point pattern with no retraining, and show they reproduce a gold-standard NUTS analysis (parameters and intensity map) in seconds rather than minutes (Section 10).

All code and experiments are released to reproduce every number and figure. Table 1 summarizes the experiments.

2 Background

Log-Gaussian Cox processes. A Cox process on a domain $D \subset \mathbb{R}^2$ is a Poisson process with a random intensity $\Lambda(s)$. In an LGCP, $\log \Lambda(s) = Z(s)$ is a Gaussian process, $Z \sim \mathcal{GP}(\mu, k)$ [Møller et al., 1998]. Discretizing D into a regular $G \times G$ grid of cells with centroids s_i and equal area a, the standard computational model [Diggle et al., 2013] takes

$$Z_i = \beta_0 + f_i, \quad f \sim \mathcal{N}(\mathbf{0}, K_{\boldsymbol{\theta}}), \quad K_{ij} = \sigma^2 \rho_\nu\big(\|s_i - s_j\|/\ell\big), \quad y_i \mid Z \sim \mathrm{Poisson}\big(a\, e^{Z_i}\big), \tag{1}$$

where ρ_ν is a Matérn correlation of smoothness ν (we use the exponential kernel $\nu = \frac{1}{2}$). The parameters $\boldsymbol{\theta} = (\beta_0, \sigma, \ell)$ are the baseline log-intensity (abundance), the marginal standard deviation of the log-intensity field (aggregation / over-dispersion), and the correlation length (spatial scale of aggregation). The latent vector $f \in \mathbb{R}^{G^2}$ is a nuisance that must be integrated out; this is the source of both the modelling power and the computational difficulty.

Neural posterior estimation. Given a prior $p(\boldsymbol{\theta})$ and an implicit simulator $p(\mathbf{x} \mid \boldsymbol{\theta})$, NPE trains a conditional density estimator by minimizing the forward Kullback–Leibler divergence,

$$\phi^\star = \arg\min_\phi \; \mathbb{E}_{\boldsymbol{\theta}\sim p(\boldsymbol{\theta})}\, \mathbb{E}_{\mathbf{x}\sim p(\mathbf{x}\mid\boldsymbol{\theta})}\big[-\log q_\phi(\boldsymbol{\theta} \mid \mathbf{x})\big], \tag{2}$$

whose minimizer is the true posterior $p(\boldsymbol{\theta} \mid \mathbf{x})$ when the family is flexible enough [Papamakarios and Murray, 2016, Greenberg et al., 2019]. We use a single-round (fully amortized) estimator so that, after training, inference on a new dataset is one forward pass. The density family is a masked autoregressive flow (MAF) [Papamakarios et al., 2017], and the conditioning $\mathbf{x}$ is a learned summary of the count grid.

3 Method

Simulator and prior. We work on the unit square discretized into a 16×16 grid (256 cells). The prior is $\beta_0 \sim \mathcal{N}(5, 0.7^2)$, $\sigma \sim \mathrm{U}(0.2, 1.8)$, $\ell \sim \mathrm{U}(0.05, 0.5)$, chosen so that prior-predictive patterns span sparse-to-dense and weakly-to-strongly aggregated regimes (Figure 1). Simulating from (1) requires, per parameter draw, a Cholesky factor of the 256×256 covariance; we batch these on the CPU so that $60,000$ datasets are produced in about a minute.

Parameter transform. So that posterior samples always respect the (bounded) prior support, the flow operates on an unconstrained, standardized reparameterization: β_0 is z-scored and σ, ℓ are mapped through a logit of their rescaled uniform range and standardized. Densities and samples are transformed back to natural space with the exact change of variables.

Summary network. The count grid is passed through a small convolutional network (two 3×3 conv blocks, average pooling, two further conv blocks, global pooling) whose output is concatenated with two explicit global statistics (log total count and log count-variance) and projected to a 48-dimensional embedding. The convolutions exploit the spatial structure of the pattern, while the explicit total-count feature gives the network direct access to the strongest signal for abundance.

Density estimator. The embedding conditions a MAF with five autoregressive transforms (each a conditional MADE with two hidden layers of width 64 and tanh-bounded log-scales for stability), with the variable order reversed between transforms. The whole model (104,398 parameters) is trained end-to-end by minimizing (2) on the $60,000$ simulations with Adam, early stopping on a validation split; training takes 22.1 minutes on four CPU cores. At test time we draw $1,000$ posterior samples per dataset.

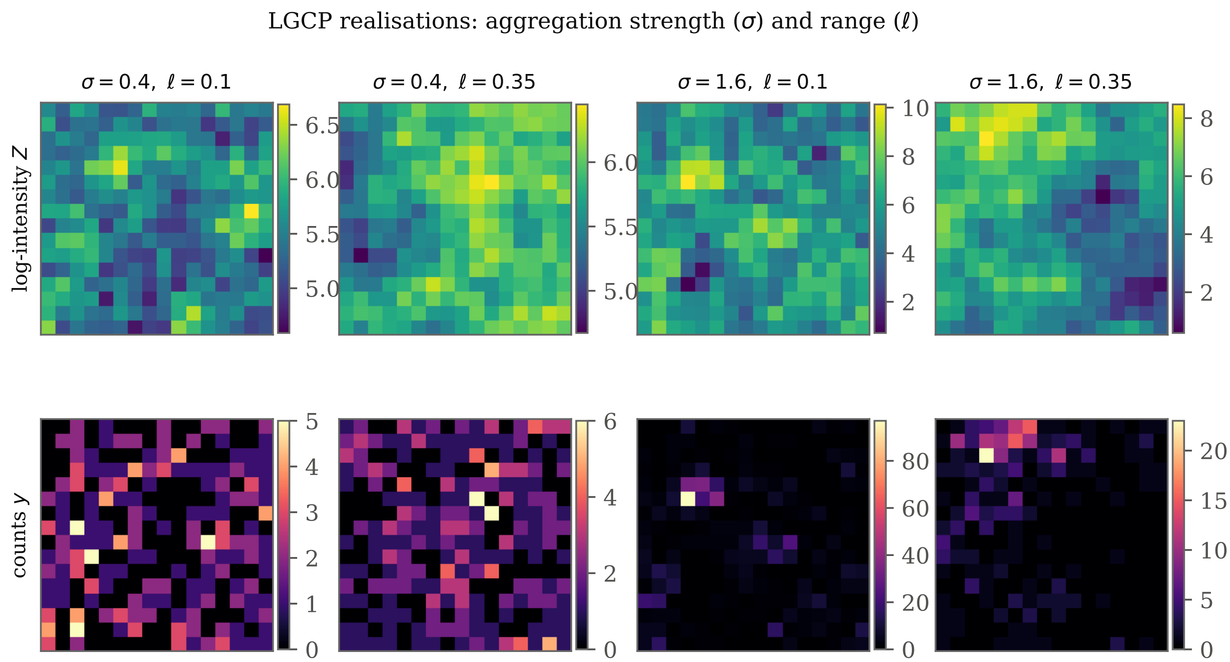

Figure 1: Prior-predictive LGCP realisations on the 16×16 grid. Top: latent log-intensity fields Z; bottom: the observed Poisson counts y. Increasing σ (left→right pairs) strengthens aggregation; increasing ℓ enlarges the spatial scale of that aggregation. Inference must recover these parameters from a *single* count grid.

Baselines. We compare against (i) an otherwise identical NPE that consumes *hand-crafted* spatial summaries—global count statistics, the index of dispersion, and a binned spatial autocorrelation profile of the log-count field—instead of the learned CNN embedding; and (ii) a *point-estimate CNN regressor* trained by mean-squared error, representing the common practice of "learning to invert" a simulator. To place the regressor on the same calibration axes we equip it with a homoscedastic Gaussian predictive whose scale is its training residual standard deviation.

Gold-standard reference. As a reference posterior we run the No-U-Turn Sampler (NUTS) [Hoffman and Gelman, 2014] on the *same* model and priors, using a non-centred parameterization $f = L_{\boldsymbol{\theta}}\,\eta$, $\eta \sim \mathcal{N}(\mathbf{0}, I)$, where the Cholesky factor $L_{\boldsymbol{\theta}}$ is rebuilt at every leapfrog step because it depends on the sampled length-scale ℓ. This per-step re-factorization is exactly the cost that amortization avoids. We run 2 chains of 600+500 iterations and monitor split-$\hat{R}$.

4 Calibration and recovery

We evaluate on $3,000$ held-out datasets drawn from the prior predictive.

Simulation-based calibration. SBC [Talts et al., 2018] exploits the fact that, if q_ϕ equals the true posterior, the rank of each true parameter within its posterior sample is uniform. Figure 2 shows the NPE rank empirical CDFs for all three parameters lying inside the 95% uniformity band, with χ^2 uniformity p-values 0.64/0.15/0.11 (all > 0.05): the amortized posterior is calibrated in the strong distributional sense SBC demands. The point-estimate regressor's ranks (orange) depart

Table 2: Held-out accuracy and calibration ($3,000$ datasets). Point-estimate accuracy (RMSE) and marginal 90% coverage are similar across methods; the minimum SBC p-value (over the three parameters) separates them. The CNN+MAF NPE passes on all parameters ($\min p > 0.05$); the hand-crafted-summary NPE is close but mildly miscalibrated; the regressor fails decisively ($\min p \approx 0$).

	RMSE ↓			90% coverage			SBC
Method	β_0	σ	ℓ	β_0	σ	ℓ	$\min p$
NPE (CNN+MAF)	0.379	0.190	0.103	0.90	0.90	0.89	0.114
NPE (hand-crafted)	0.399	0.200	0.107	0.89	0.90	0.89	0.009
CNN regressor	0.379	0.192	0.104	0.89	0.92	0.90	0.000

drastically from uniform ($p < 0.01$ on every parameter), the signature of a posterior approximation that is systematically wrong per dataset.

Marginal coverage. Figure 3 plots empirical against nominal central-interval coverage. Both NPE variants track the diagonal closely—the CNN+MAF model's 90% intervals cover 0.90/0.90/0.89 for $\beta_0/\sigma/\ell$. Strikingly, the regressor *also* appears near-nominal here (0.89/0.92/0.90). This is not a contradiction of its failed SBC: a homoscedastic predictive whose width is tuned on held-out residuals is guaranteed to cover correctly *on average over the prior*. Marginal coverage is necessary but not sufficient—and on its own it would have hidden the regressor's defect.

Adaptive uncertainty. SBC's verdict is made concrete by asking whether the reported uncertainty predicts the actual error. Figure 4 bins datasets by their reported posterior standard deviation and plots the realised RMSE in each bin. For the NPE the two rise together along the diagonal—the posterior is *sharper* on easy datasets and wider on hard ones—so its width is informative (correlation between reported SD and error 0.37/0.40/0.43 for $\beta_0/\sigma/\ell$). The regressor's homoscedastic predictive cannot do this: most starkly for β_0, whose intervals have a nearly constant width (coefficient of variation 0.02 versus 0.30 for the NPE) and zero correlation with the error ($r = 0.01$). This is why the regressor fails SBC while looking calibrated marginally, and it is the central practical warning of the paper: inverting a simulator with a regression network yields uncertainty that is uninformative per dataset, whereas NPE yields a calibrated, input-adaptive posterior at essentially the same training cost.

Recovery. Figure 5 shows posterior means against truth. Abundance β_0 and aggregation σ are recovered well; the correlation length ℓ is intrinsically harder to pin down from a single realization, and— importantly—the NPE *reports* this through wider, still-calibrated intervals rather than confident errors. Coefficients of determination are 0.70/0.83/0.37. The weaker identifiability of β_0 against the field mean (a known feature of LGCPs) is faithfully reflected in the posterior width, not hidden.

5 Agreement with gold-standard MCMC

Calibration shows the posterior is right *on average* over the prior; we now ask whether it is right *for individual datasets* by comparing, dataset by dataset, against NUTS. We ran NUTS on 30 test datasets that reached convergence ($\hat{R} < 1.1$).

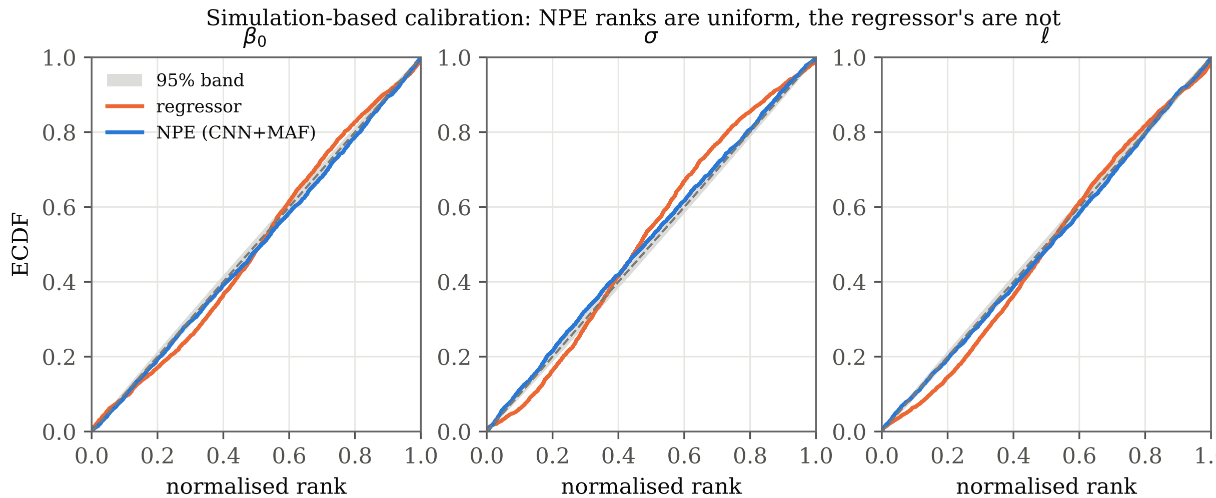

Figure 2: Simulation-based calibration. NPE rank ECDFs (blue) for the three parameters fall within the 95% uniformity band (grey); the diagonal is perfect calibration. The point-estimate regressor (orange) departs sharply from uniform, revealing per-dataset miscalibration.

Table 3: Per-parameter agreement between NPE and NUTS over 30 datasets. C2ST is computed on the joint 3-D posteriors (0.5 = indistinguishable).

Parameter	C2ST	1-Wass. (/prior sd)	mean corr	sd ratio
β_0	–	0.056	0.998	1.03
σ	–	0.053	0.998	1.04
ℓ	–	0.091	0.989	1.03
overall (3-D)	0.543	0.067	–	–

Figure 6 tells the story. The overlaid marginal posteriors (top) are nearly coincident with the NUTS reference for a representative dataset. The posterior means agree tightly (correlation ≥ 0.99 across parameters) and, crucially, so do the posterior *standard deviations*—the NPE is not merely locating the posterior but reproducing its spread. Aggregated over datasets, a classifier two-sample test cannot tell NPE from NUTS samples apart: mean C2ST accuracy is 0.543 against a chance level of 0.5, and the mean marginal 1-Wasserstein distance is only 0.067 prior standard deviations. Table 3 collects these numbers.

The practical pay-off is in Figure 7. A single NUTS fit takes 280 s; an amortized NPE posterior takes 12.67 ms, a $\sim 22,123\times$ per-dataset speed-up. Accounting for the one-time 22.1-minute training cost, NPE overtakes repeated MCMC after just 5 datasets—a threshold trivially exceeded in any realistic multi-region or multi-species study.

6 From parameters to the intensity surface

So far, as in prior amortized-LGCP work [Wang et al., 2026], the target has been the global parameters. But in most applications—disease mapping, species distribution, hotspot detection—the quantity of scientific interest is the *surface* itself: the latent log-intensity field $Z(s) = \beta_0 + f(s)$, or the intensity $\Lambda(s) = e^{Z(s)}$, at every location, with uncertainty. We show that this map, too, can

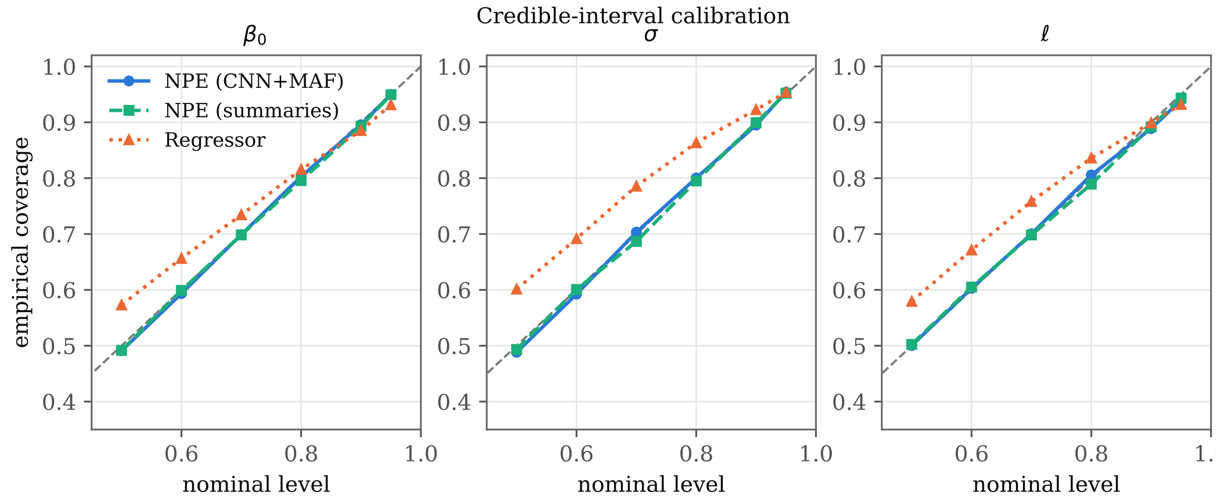

Figure 3: Marginal credible-interval calibration. All three methods track the diagonal: the regressor's tuned homoscedastic width makes it look calibrated *marginally*, which Figure 4 shows to be misleading.

be produced by a single amortized forward pass.

An amortized field posterior. We train a U-Net that maps the count grid to a *structured Gaussian* posterior over the whole field,

$$Z \mid y \sim \mathcal{N}\big(\mu(y),\ \mathrm{diag}(d(y)^2) + V(y)V(y)^\top\big), \tag{3}$$

where $\mu \in \mathbb{R}^{G^2}$ is the posterior mean map, d the per-cell scale, and $V \in \mathbb{R}^{G^2\times k}$ a rank-k factor ($k = 16$). The network is trained by maximizing the density (3) of the true field under the same simulator, using the Woodbury identity so the $G^2 \times G^2$ covariance is never formed. The mean is parameterized as a data-anchored residual around the pointwise estimate $\log((y_i + \frac{1}{2})/a)$, so the surface has the correct absolute scale and the network learns only the spatial correction and the uncertainty. Training takes 200.4 min; thereafter each map—mean and full joint uncertainty—is one forward pass.

Why the low-rank term matters. The field posterior is spatially *correlated*: neighbouring cells are uncertain together. A purely per-cell (diagonal) posterior can be calibrated pointwise yet badly misjudge *aggregate* functionals—a regional case count or the total abundance $N = \int \Lambda$—whose error variance is dominated by those correlations. The low-rank factor V captures them, at negligible cost, and is what makes the aggregates calibrated.

Results. Figure 8 shows reconstructions: the posterior mean recovers the true surface (test-set correlation 0.893, RMSE 0.59 on the log scale) and the posterior SD map correctly localizes uncertainty to the low-count regions where the field is genuinely unconstrained. Figure 9 quantifies calibration. Pointwise intervals are near-nominal (90% coverage 0.90) and the pixelwise z-scores are standard normal (SD 1.01). For total abundance, the full posterior is calibrated (90% coverage 0.93) while the diagonal-only ablation collapses to 0.56—concrete evidence that the low-rank correlation is necessary, not decorative; the same holds for regional counts (0.89 vs. 0.59). Finally, against a gold-standard NUTS field posterior (Figure 10), the amortized posterior agrees closely: over

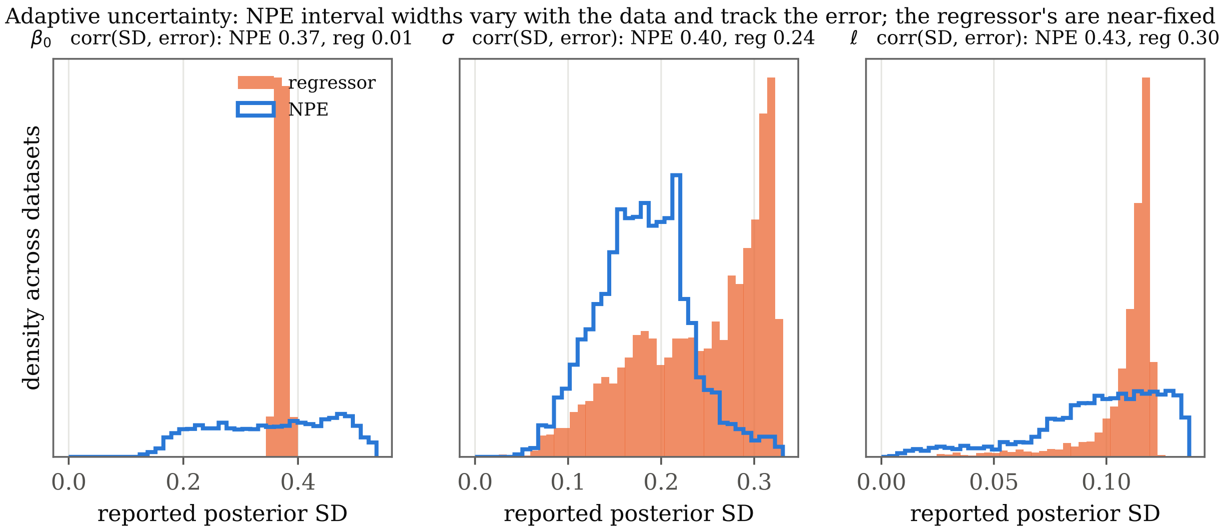

Figure 4: Does the reported uncertainty predict the error? Datasets are binned by their posterior standard deviation (x); y is the realised RMSE in each bin. NPE (blue) follows the diagonal—wider intervals where the error is genuinely larger—so its uncertainty is informative. The regressor (orange) is flat, especially for β_0: its fixed-width predictive says nothing about which estimates are reliable. This is the per-dataset miscalibration that SBC detects and marginal coverage hides.

12 datasets the posterior-mean maps correlate at 0.990 and the posterior-SD maps at 0.84, with matching pointwise coverage (0.91 vs. 0.91). Amortized inference thus delivers not only calibrated parameters but a calibrated *map*—the object practitioners actually act on.

7 Robustness to misspecification

A method that is only validated in-distribution can fail silently when the world does not match the simulator. We therefore train on the exponential ($\nu = \frac{1}{2}$) kernel but *test* on data generated from rougher-to-smoother Matérn kernels ($\nu = \frac{3}{2}, \frac{5}{2}$), a realistic form of model mismatch since the true smoothness is rarely known. Figure 11 shows coverage degrading gracefully: intervals remain informative and only mildly miscalibrated under moderate mismatch (at $\nu = \frac{5}{2}$, 90% coverage 0.87/0.75/0.75), with ℓ—whose meaning is most kernel-dependent —the most affected. The failure mode is gradual and diagnosable rather than catastrophic, and it is detectable in practice with the same SBC/coverage tools used above, supporting a workflow in which the simulator's smoothness is itself treated as uncertain.

8 Amortized inference under preferential sampling

The results so far establish that amortized NPE is a *trustworthy* drop-in for MCMC on the standard, fully observed LGCP—the setting of prior amortized work [Wang et al., 2026]. We now turn the argument around and use amortization to do something the standard toolkit finds genuinely hard, showing that the likelihood-free nature of SBI is not just a convenience but a modelling capability.

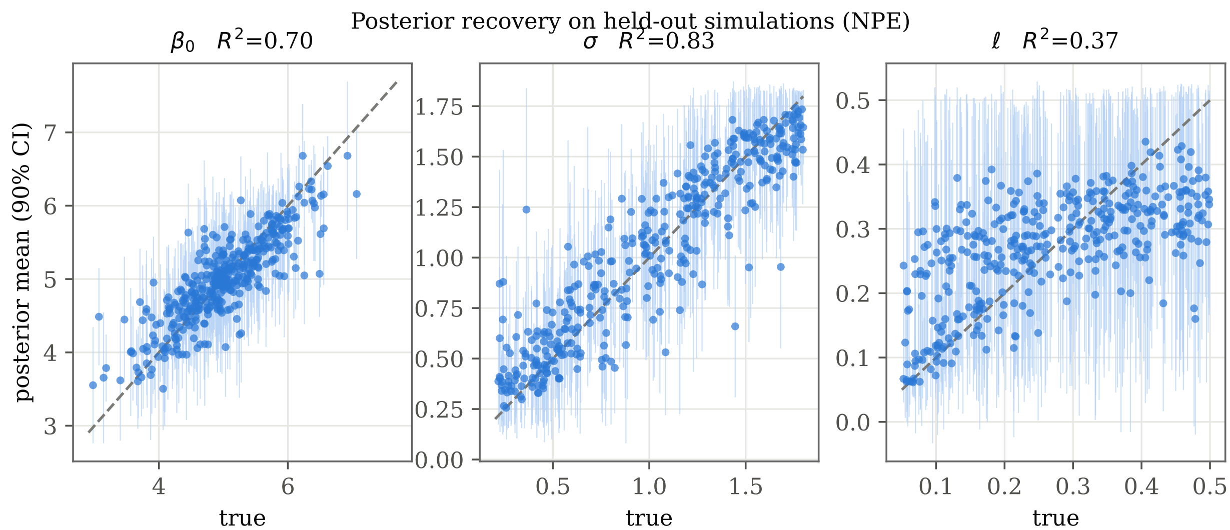

Figure 5: Posterior recovery on held-out simulations. Points are posterior means with 90% credible intervals against the true parameter; the dashed line is $y = x$.

The problem. Classical LGCP inference assumes the observation locations are chosen independently of the latent field—an *ignorable* design. Frequently they are not: monitors are sited where pollution is expected, sightings are logged where animals are abundant, tissue is sampled where a biofilm looks dense. Under such *preferential sampling* [Diggle et al., 2010], the observed locations carry information about the field, and an analysis that ignores this is biased—typically overstating abundance and variability, because one preferentially sees high-intensity locations. Correcting it with likelihood-based tools requires building and fitting a bespoke *joint* model of the field, the counts, and the sampling mechanism, redone for every design.

An amortized, likelihood-free treatment. For SBI the correction is almost free: the design is simply part of the simulator. We augment the generator with a preferential observation model in which cell i is observed with probability $\pi_i = \mathrm{sigmoid}(\gamma f_i)$, where $f_i = Z_i - \beta_0$ is the mean-zero field and $\gamma \geq 0$ controls the strength of preferentiality ($\gamma = 0$ recovers a uniform design). The data are the observed counts together with the observation mask, and the unknowns are now $(\beta_0, \sigma, \ell, \gamma)$. We train a *design-aware* NPE on this generator—a two-channel convolutional summary (masked log-counts and the mask itself) feeding the flow—and, for contrast, a *design-ignorant* NPE that embodies the standard assumption: it is trained on uniform-design simulations and applied, as a practitioner would, to preferentially sampled data. No new derivation is needed for either; only the simulator changes.

Results. Figure 12 and Table 4 tell the story on $3,000$ preferentially sampled test datasets. The design-ignorant analysis is biased, and increasingly so as preferentiality grows: it overestimates the baseline intensity (bias $+0.23$ in β_0) and underestimates the spatial variance (bias -0.11 in σ, because it sees only the high-intensity part of the field), with both biases rising monotonically in γ and its 90% intervals under-covering (down to 0.81; SBC $p < 0.001$ on β_0 and σ). The design-aware NPE is essentially unbiased across the whole range of γ (bias $+0.00$ and $+0.00$) and restores coverage to nominal (0.89 at worst), with SBC satisfied on exactly the parameters the design corrupts ($p = 0.07$ and 0.06 for β_0, σ; the harder-to-identify range ℓ shows a mild residual

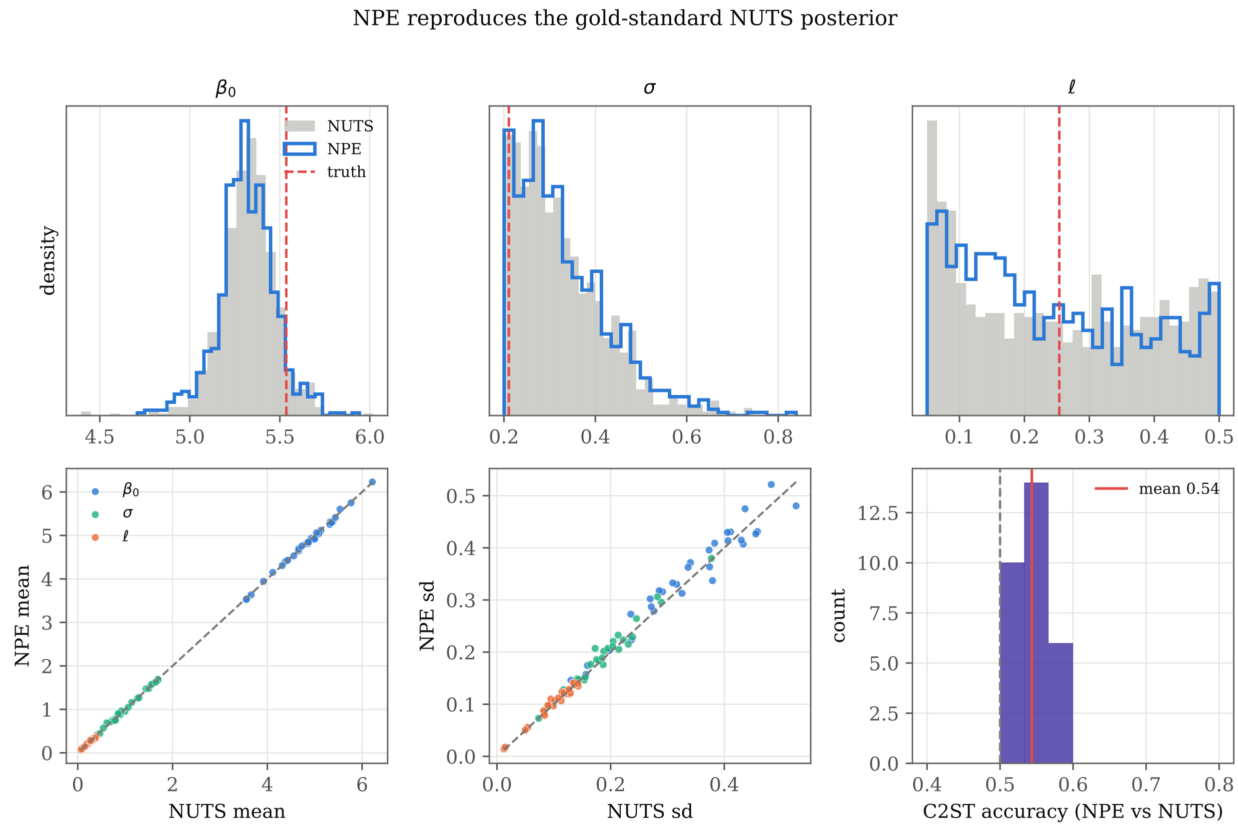

Figure 6: NPE versus gold-standard NUTS. *Top:* overlaid marginal posteriors for one held-out dataset (NPE step outline, NUTS filled, truth dashed). *Bottom:* across datasets, posterior means and standard deviations agree with NUTS, and the classifier two-sample accuracy concentrates near the chance level 0.5 (indistinguishable posteriors).

SBC deviation despite its nominal coverage). As a bonus the ignorable analysis cannot provide, it also infers the preferentiality itself (γ recovered with $R^2 = 0.68$; Figure 13). A gold-standard NUTS sampler for the joint preferential model confirms the design-aware posterior closely tracks the true one, not merely a self-consistent approximation: over 12 datasets the marginals agree to within 0.14 prior standard deviations (mean 1-Wasserstein), and a classifier separates the two posteriors only modestly above chance (C2ST 0.64 versus 0.5 for identical posteriors). The small residual—larger here than for the fully observed model (Section 5)—is consistent with the mild SBC deviations on the harder parameters, and is the price of the more challenging four-parameter, partially observed inference; the first-order conclusions (bias removal and restored coverage) are unaffected.

Why this matters. Preferential sampling is one instance of a general pattern. Thinning, detection error, aggregation to coarse regions, and preferential or adaptive designs all change the *observation* model while leaving the latent LGCP intact, and all break the clean likelihoods that INLA-style methods rely on. For amortized SBI each is a change to a few lines of the simulator, after which inference is not only fast but—as we have verified here—calibrated and, where the standard analysis is biased, corrected. This is the methodological payoff of trustworthy amortization.

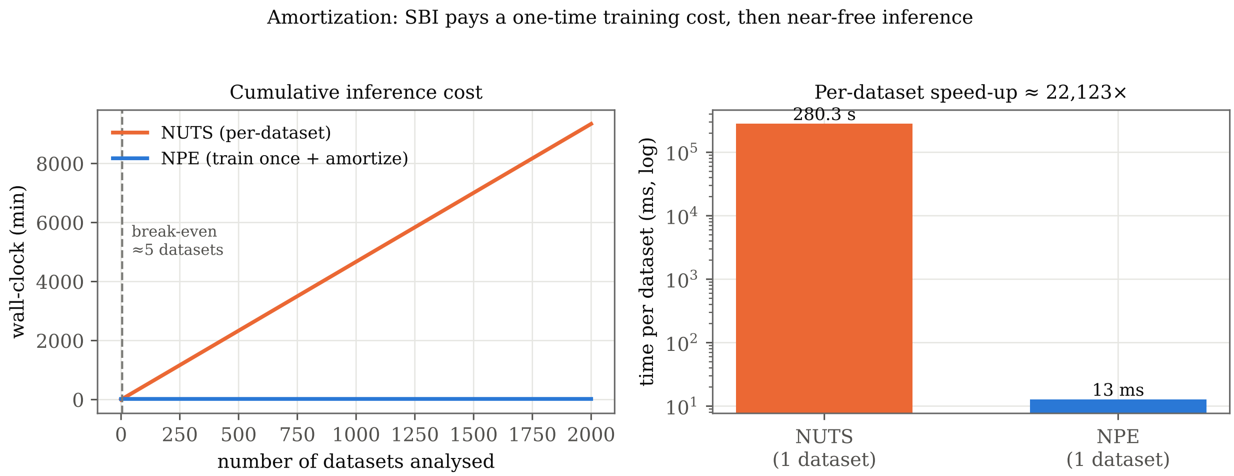

Figure 7: Amortization. *Left:* cumulative wall-clock against number of datasets; the NPE line (training cost + near-free inference) crosses the per-dataset MCMC line after 5 datasets. *Right:* per-dataset cost on a log scale.

Table 4: Inference under preferential sampling ($3,000$ test datasets). Ignoring the informative design biases β_0 and σ and breaks calibration; the design-aware NPE is unbiased and calibrated.

	bias			90% coverage			SBC p	
Analysis	β_0	σ	ℓ	β_0	σ	ℓ	β_0	σ
Design-ignorant	+0.23	-0.11	-0.00	0.82	0.81	0.91	0.000	0.000
Design-aware (ours)	+0.00	+0.00	-0.01	0.89	0.90	0.90	0.07	0.055

9 Change of support: inference from aggregated counts

Preferential sampling is one non-standard observation model; spatial *aggregation* is another, and arguably the more pervasive one. In spatial epidemiology and the social sciences the point pattern is almost never observed directly—counts are released as totals over coarse administrative regions. Recovering fine-scale structure from such totals is the change-of-support (or spatial-misalignment) problem [Gotway and Young, 2002]. Its likelihood is awkward for analytic tools because each region's contribution is a sum of correlated log-Gaussian Poisson rates; for amortized SBI it is, once more, just a change to the simulator—aggregate the simulated fine counts to the observation support and learn the posterior from the coarse totals (Figure 14).

Setup and results. We observe totals over a 4×4 partition of the 16×16 grid—a sixteen-fold aggregation—and infer (β_0, σ, ℓ) from the sixteen regional counts with a convolutional NPE trained on the aggregated simulator. Two things stand out (Figure 15). First, the posterior remains *calibrated* despite the loss of information: SBC holds ($\min p = 0.09$) and 90% coverage is near-nominal (0.89/0.90/0.90). Second, the method is *honest about what aggregation destroys.* Baseline intensity is still well recovered (R^2 falls only from 0.70 to 0.68), but the correlation range is largely washed out—its R^2 drops from 0.37 to 0.17 because within-region spatial variation is exactly what aggregation removes—and the posteriors widen accordingly (1.2× wider for ℓ versus the fully observed case, against only 1.1× for β_0). The uncertainty grows to match the information

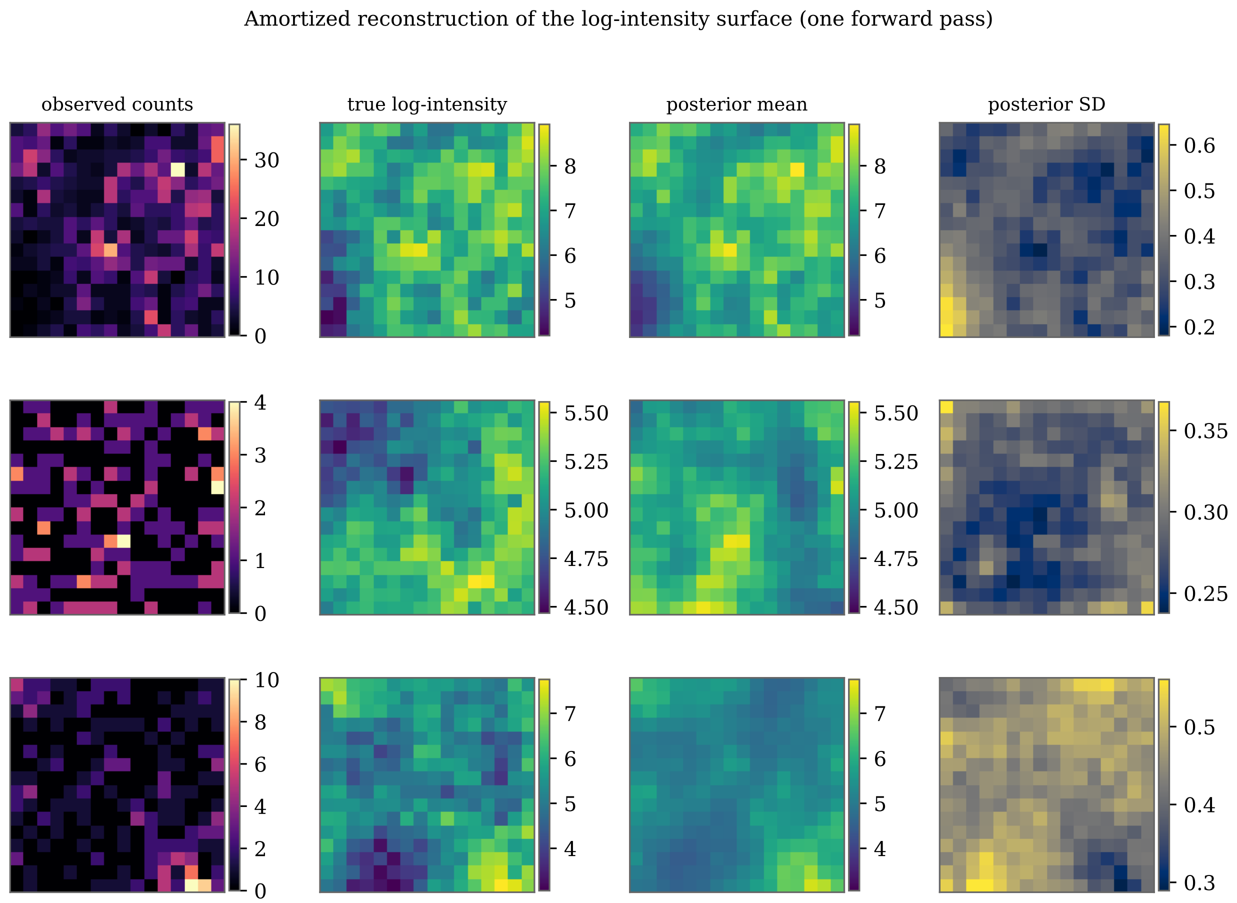

Figure 8: Amortized intensity-surface reconstruction for three held-out patterns. From the counts (left) a single forward pass returns the posterior mean log-intensity (matching the truth) and a posterior-SD map that concentrates uncertainty in the sparsely observed regions.

lost, rather than staying falsely confident. A gold-standard NUTS sampler for the aggregated model confirms the amortized posterior is correct (C2ST 0.52 over 11 datasets), at 529 s per NUTS fit versus milliseconds for the amortized estimator.

10 A real-data case study

Everything so far is simulated, which is what makes calibration and gold-standard comparison possible. We close with a real point pattern to show the pipeline transfers. We use the `crimes` pattern from the geodanet example dataset distributed with the PySAL library [Rey and Anselin, 2007]: 287 recorded crime locations in a road-network neighbourhood, binned to the same 16×16 grid. Crucially we *do not retrain*: the very same NPE and field U-Net used throughout, trained only on synthetic LGCPs, are applied to the real grid in a single forward pass each—the amortization dividend in its purest form.

Figure 16 shows the result. The amortized estimator returns a parameter posterior and a full posterior intensity *map* with uncertainty, and both agree with a gold-standard NUTS analysis of the same data: the parameter posteriors overlap (C2ST 0.57; posterior means $\beta_0 = 4.86$ vs. 4.87, $\sigma = 1.42$ vs. 1.39, $\ell = 0.06$ vs. 0.06 for NPE vs. NUTS), and the posterior-mean intensity maps

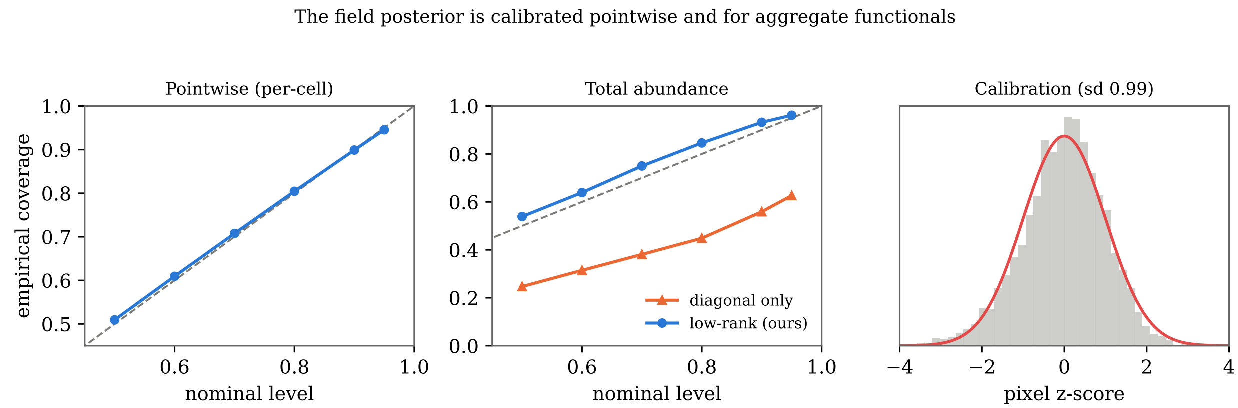

Figure 9: Field-posterior calibration. *Left:* near-nominal pointwise coverage. *Middle:* total-abundance coverage—the full low-rank posterior is calibrated, the diagonal-only ablation is badly overconfident. *Right:* pixelwise z-scores match $\mathcal{N}(0, 1)$.

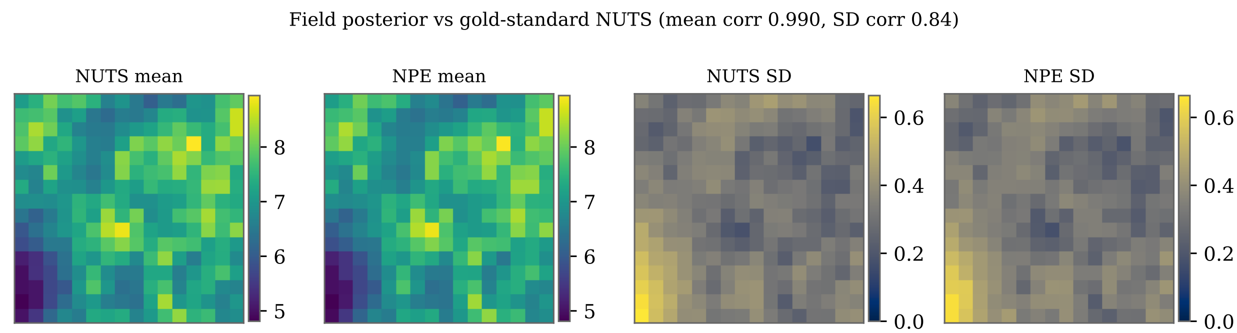

Figure 10: The amortized field posterior (mean and SD maps) against gold-standard NUTS for an example pattern; the two are visually and quantitatively close.

correlate at 0.993 (SD maps at 0.96). The amortized analysis takes 2313 ms; NUTS takes 811 s. This is a validation of the amortized machinery, not a claim that a stationary LGCP is the true generative model for network-constrained crime—indeed the mild model mismatch is exactly the regime Section 7 anticipates—but it demonstrates that a model trained entirely in simulation delivers calibrated, MCMC-quality inference on real data instantly.

11 Related work

Simulation-based inference. NPE originates with mixture-density and flow-based conditional estimators [Papamakarios and Murray, 2016], refined by sequential and automatic-posterior variants [Lueckmann et al., 2017, Greenberg et al., 2019] and surveyed by Cranmer et al. [2020]. Alternatives estimate the likelihood [Papamakarios et al., 2019] or the likelihood ratio [Hermans et al., 2020]; Lueckmann et al. [2021] benchmark them. Our density estimator is a masked autoregressive flow [Papamakarios et al., 2017, Rezende and Mohamed, 2015]. Calibration is assessed with simulation-based calibration [Talts et al., 2018] and the classifier two-sample test [Lopez-Paz and Oquab, 2017]; recent work stresses that SBI must be checked, not assumed [Hermans et al., 2022].

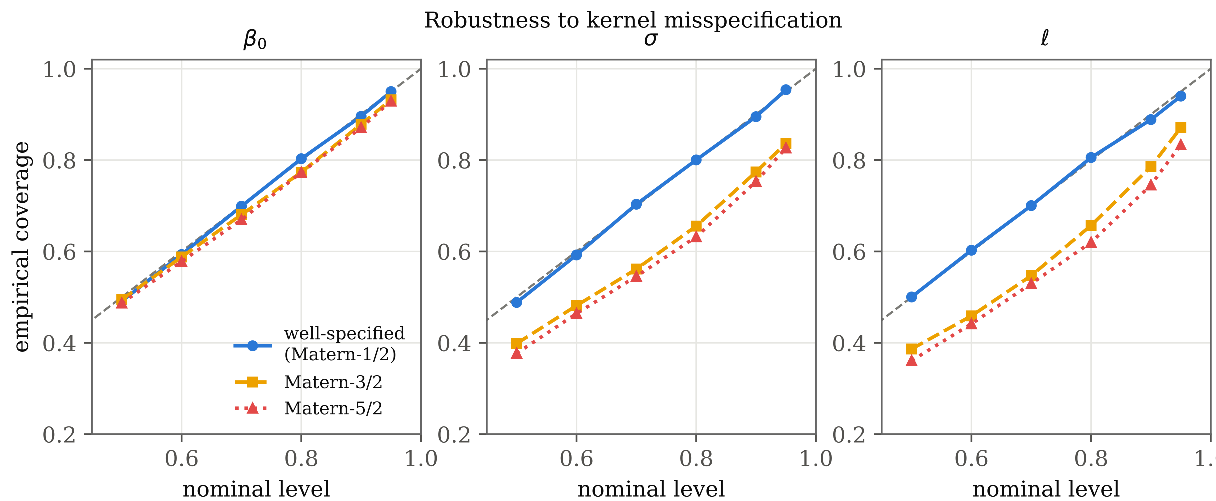

Figure 11: Kernel misspecification. Coverage of the NPE (trained on $\nu = \frac{1}{2}$) under test data from $\nu = \frac{1}{2}$ (well-specified), $\frac{3}{2}$ and $\frac{5}{2}$. Degradation is gradual and largest for the range ℓ.

Inference for LGCPs and spatial models. Classical LGCP inference uses MCMC over the latent field [Møller et al., 1998, Christensen et al., 2006], with elliptical slice sampling and manifold methods to cope with the field–hyperparameter coupling [Murray et al., 2010, Girolami and Calderhead, 2011]. INLA with an SPDE representation [Rue et al., 2009, Lindgren et al., 2011, Simpson et al., 2016] is the dominant fast, approximate alternative. These are all *per-dataset* solvers; our contribution is orthogonal—an amortized posterior shared across datasets—and complementary, since INLA or NUTS can serve as the gold standard we validate against.

Neural inference for spatial and point-process models. Amortized and likelihood-free neural inference has moved rapidly into spatial statistics and ecology; see Zammit-Mangion et al. [2025] for a review. Two threads are directly relevant. The first estimates parameters as *point predictions*: Vihrs [2022] trains neural networks to recover spatial point-process parameters from a pattern, and neural Bayes estimators provide fast, amortized point estimators for spatial covariance and extremal models, with bootstrap uncertainty [Gerber and Nychka, 2021, Lenzi et al., 2023, Sainsbury-Dale et al., 2024, 2025]. Our regressor baseline is a representative of this thread, and our calibration analysis quantifies precisely what its uncertainty does and does not deliver. The second thread learns *full posteriors*: closest to us, Wang et al. [2026] apply BayesFlow [Radev et al., 2020] to one- and two-dimensional LGCPs and validate against MCMC on selected realizations, with an oral-microbiome application. We are complementary to that work: rather than a new application, we contribute a systematic *trustworthiness audit*—calibration across the entire prior via SBC, a controlled comparison against the point-estimate paradigm, a summary-statistic ablation, a classifier-two-sample and Wasserstein benchmark against MCMC in two dimensions, and a kernel-misspecification stress test—using a deliberately minimal, from-scratch estimator so that the conclusions are about the approach, not a particular software stack.

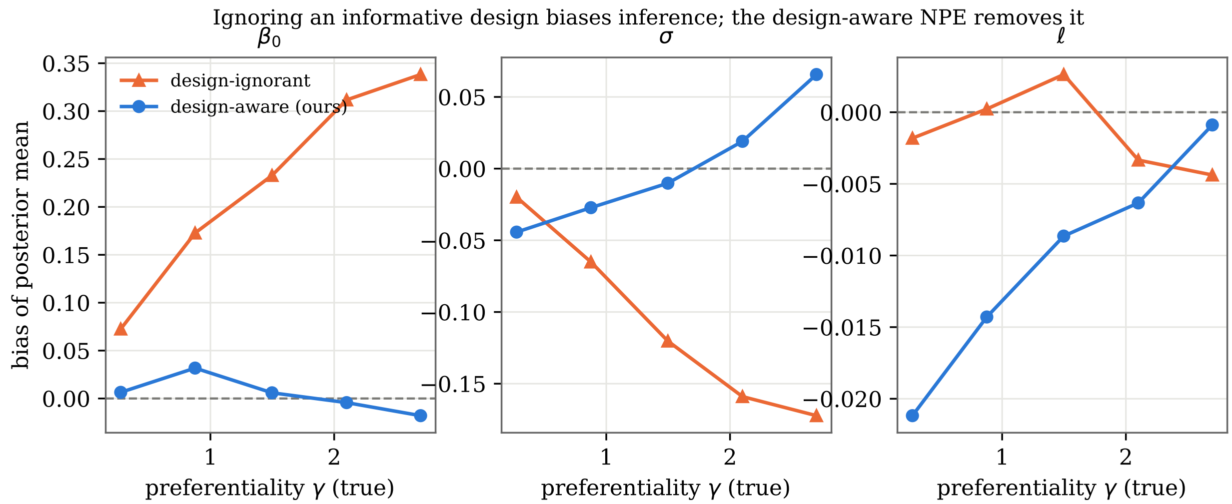

Figure 12: Bias of the posterior mean under preferential sampling, against the true preferentiality γ. The design-ignorant analysis (orange) is increasingly biased in β_0 and σ as sampling becomes more informative; the design-aware NPE (blue) stays essentially unbiased. Preferential sampling chiefly distorts level and variance—note the much finer vertical scale for the range ℓ, where both methods are effectively unbiased.

12 Discussion

We have shown that amortized NPE gives calibrated posteriors for LGCP parameters that match gold-standard MCMC at a fraction of the per-dataset cost, and that the common regression shortcut, while accurate in the mean, is dangerously overconfident. The recipe is simple and dependency-light: simulate, embed with a CNN, fit a flow, and—critically—*check* with SBC, coverage and a gold-standard comparison before trusting the result.

Limitations and outlook. Our study fixes the grid resolution and the kernel family; the misspecification analysis shows real sensitivity to smoothness, and the field posterior, though calibrated, uses a low-rank Gaussian form that may understate strongly non-Gaussian tails. Two directions we have begun here seem the most consequential. The intensity-surface posterior (Section 6) turns amortized inference into amortized *mapping*, the object practitioners act on; richer posterior families (conditional flows or diffusions over the field) and joint inference with the hyperparameters are natural refinements. The preferential-sampling and change-of-support results (Sections 8–9) show that once amortized inference is *trusted*, the likelihood-free interface makes non-standard observation models tractable almost for free; thinning and imperfect detection, and adaptive designs are immediate further members of that family. The most important gap is external validity—all our experiments are simulated; a real application, where the generative model is itself uncertain, is the essential next step. Amortizing over grid resolution and domain shape, and treating the kernel smoothness as an inferred parameter to convert the misspecification sensitivity into robustness, round out the agenda. We view calibrated amortized inference—of parameters, surfaces, and observation processes alike—as a practical path to LGCP modelling at the scale and observational complexity that contemporary spatial data demand.

Reproducibility. The simulator, method, diagnostics and all experiment scripts are released; every figure and number in this paper is regenerated by the accompanying code.

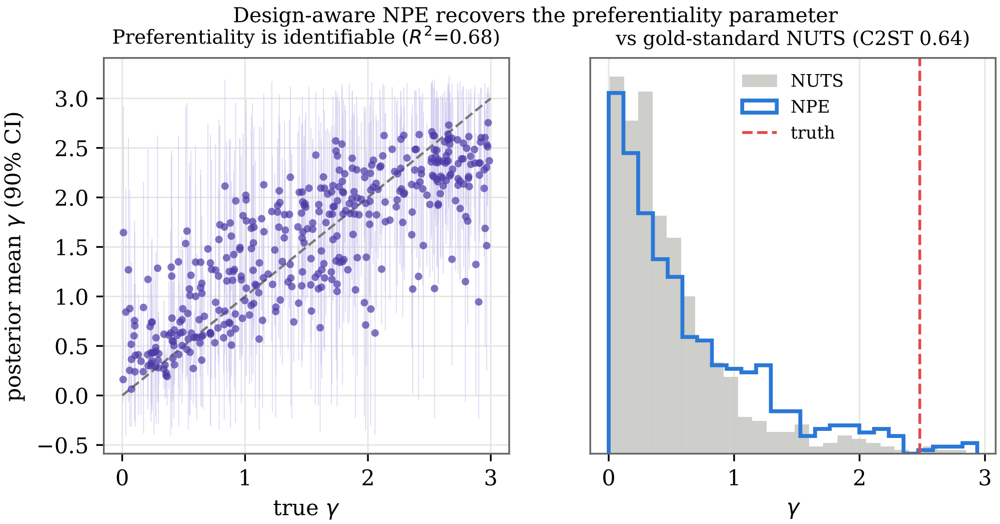

Figure 13: *Left:* the design-aware NPE identifies the preferentiality parameter γ from a single pattern. *Right:* for an example dataset its γ posterior matches gold-standard NUTS for the joint preferential model.

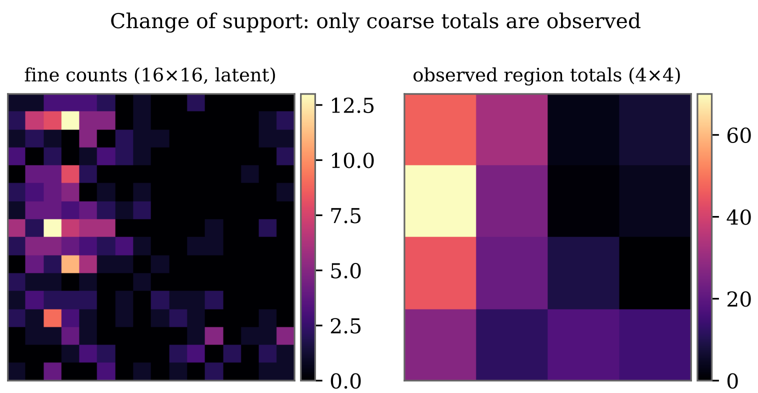

Figure 14: Change of support. The fine point pattern (left) is latent; only coarse regional totals (right) are observed. Inference must recover fine-scale parameters from the aggregated counts.

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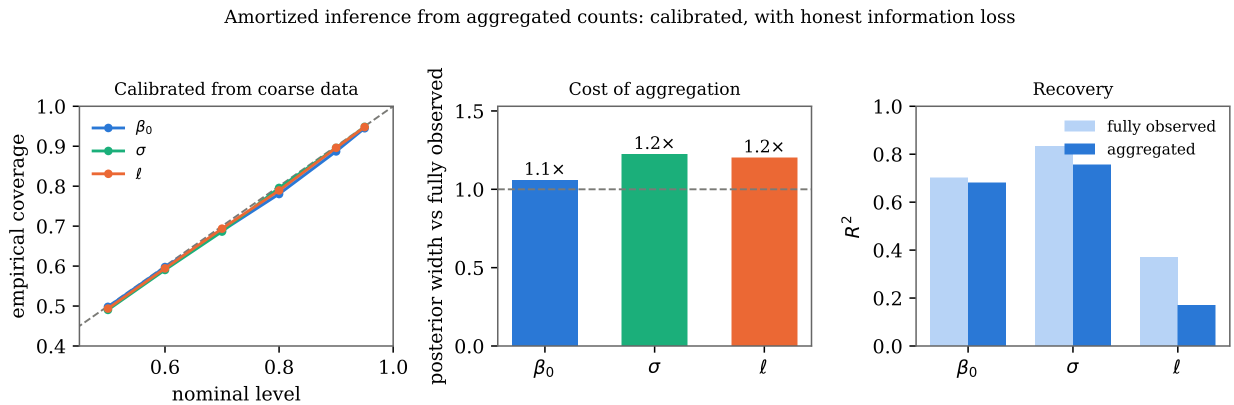

Figure 15: Inference from aggregated counts. *Left:* coverage stays near nominal. *Middle:* posteriors widen relative to the fully observed case, most for the range ℓ. *Right:* recovery (R^2) is preserved for β_0 but collapses for ℓ—aggregation removes the within-region variation that identifies it. Calibrated inference tracks the true information content.

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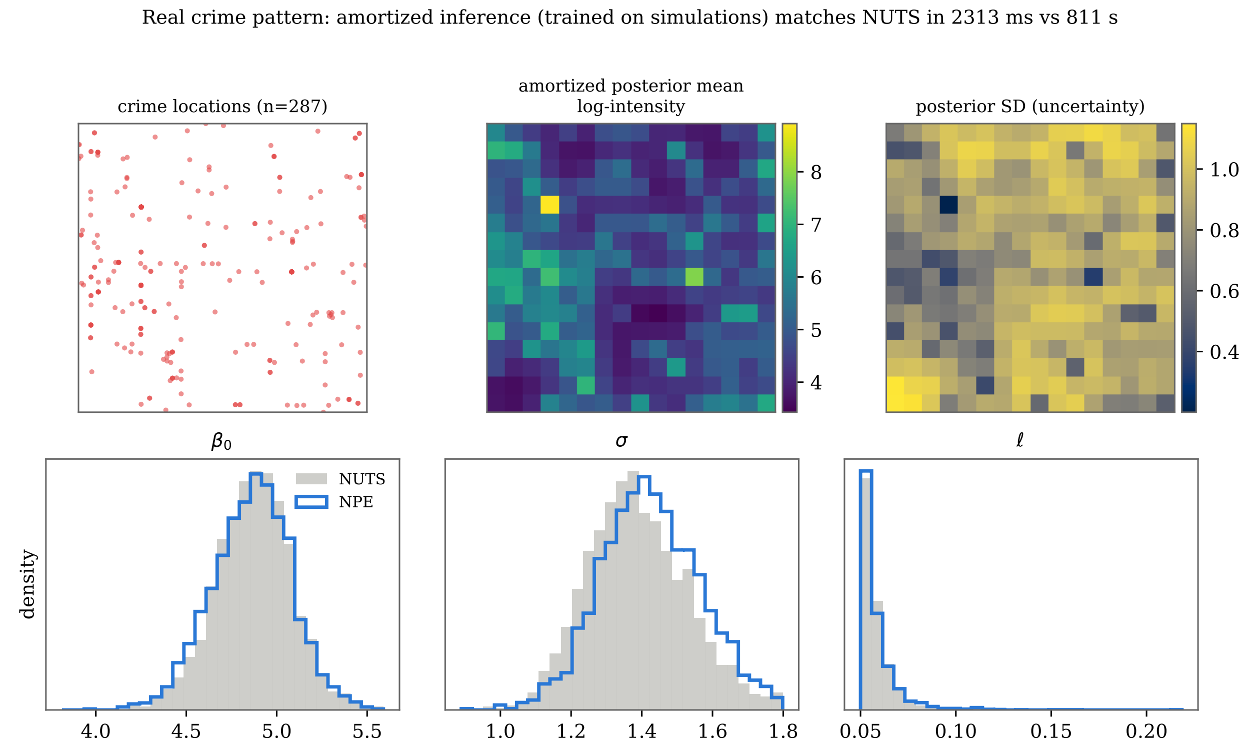

Figure 16: Real crime point pattern. *Top:* the observed events, and the amortized posterior mean and SD of the log-intensity surface (one forward pass each, from a network trained only on simulations). *Bottom:* the amortized parameter posteriors (blue) against gold-standard NUTS (grey) on the same data.

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